POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 5013 | Semester V | Theory | 4 Credits

Hydraulics & Irrigation Engineering

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Civil Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 5013 as Hydraulics & Irrigation Engineering in Semester V for Civil Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Kharagpur’s Hydraulic Engineering course and polytechnic water resource curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Hydraulic designs, pipe networks, dam structures and irrigation schedules must be based on approved engineering plans, certified hydrological data, current safety standards and competent professional review.

Verified source note: Verified sources support fluid mechanics, open channel flow, irrigation methods, and dam engineering. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Hydraulics & Irrigation Engineering studies the behavior of water at rest and in motion, and its application in agriculture through irrigation systems. It develops the ability to analyze fluid properties, evaluate flow through pipes and open channels, understand the design of dams and canals, and coordinate irrigation solutions while respecting the technical and environmental boundaries of civil engineering.

> **Malayalam support:** □ handbook □□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□ □□□□□□□□□.

Prerequisite foundation

Engineering mechanics, basic physics, mathematics and surveying.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the properties of fluids, hydrostatics and principles of fluid flow.
2. Analyze flow through orifices, mouthpieces, pipes and open channels.
3. Evaluate the methods of irrigation, crop water requirements and canal systems.
4. Explain the systematic design of dams, spillways and river training works.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□ exam-□ □□□□□ □□□□□□□□ □□□□□□ □□□□□ □□□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain fluid properties, hydrostatics and the fundamentals of fluid flow.	Understand
CO2	Analyze flow through pipes, orifices and open channels for civil works.	Apply
CO3	Evaluate irrigation methods, crop requirements and canal network designs.	Apply
CO4	Explain the design principles of dams, spillways and irrigation structures.	Understand

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7-PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Fluid Properties and Hydrostatics	Fluid properties: Density, Viscosity, Surface tension; Hydrostatic pressure: Pascal's law, Total pressure, Center of pressure; Buoyancy and Floatation; Fundamentals of fluid flow: Continuity equation, Bernoulli's theorem.	Approx. 14 hours	CO1	Module 1
2	Flow Through Pipes and Open Channels	Flow through orifices and mouthpieces; Flow through pipes: Darcy-Weisbach equation, Major and minor losses; Open channel flow: Chezy's and Manning's equations, Most economical sections.	Approx. 16 hours	CO2	Module 2
3	Irrigation Engineering and Canals	Methods of irrigation: Surface, Sub-surface, Sprinkler, Drip; Crop water requirements: Delta, Duty, Base period; Canal systems: Alignment, Lining, Cross-drainage works.	Approx. 16 hours	CO3	Module 3
4	Dams and Water Resource Management	Gravity dams: Forces, Stability, Failure modes; Spillways and Gates; River training works: Spurs, Groynes, Embankments; Rainwater harvesting and Watershed management.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Fluid Properties and Hydrostatics

Fluid properties: Density, Viscosity, Surface tension; Hydrostatic pressure: Pascal's law, Total pressure, Center of pressure; Buoyancy and Floatation; Fundamentals of fluid flow: Continuity equation, Bernoulli's theorem.

CO1 · Approx. 14 hours

Flow Through Pipes and Open Channels

Flow through orifices and mouthpieces; Flow through pipes: Darcy-Weisbach equation, Major and minor losses; Open channel flow: Chezy's and Manning's equations, Most economical sections.

CO2 · Approx. 16 hours

Irrigation Engineering and Canals

Methods of irrigation: Surface, Sub-surface, Sprinkler, Drip; Crop water requirements: Delta, Duty, Base period; Canal systems: Alignment, Lining, Cross-drainage works.

CO3 · Approx. 16 hours

Dams and Water Resource Management

Gravity dams: Forces, Stability, Failure modes; Spillways and Gates; River training works: Spurs, Groynes, Embankments; Rainwater harvesting and Watershed management.

CO4 · Approx. 14 hours

MODULE 1

Fluid Properties and Hydrostatics

Fluid properties: Density, Viscosity, Surface tension; Hydrostatic pressure: Pascal's law, Total pressure, Center of pressure; Buoyancy and Floatation; Fundamentals of fluid flow: Continuity equation, Bernoulli's theorem.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

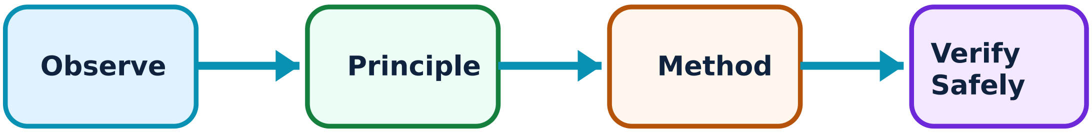
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Fluid Properties and Hydrostatics: identify the system, state its principle, follow the correct method and verify the result safely.

1. Fluid Properties and Pressure−

Fluids (liquids and gases) deform continuously under shear stress. Key properties include density, viscosity (resistance to flow), and surface tension. Hydrostatics deals with fluids at rest. Pascal's law states that pressure at any point in a fluid at rest is equal in all directions.

Study and application method

Define 'Viscosity' and 'Surface Tension'; state 'Pascal's Law' and its application in hydraulic jacks.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define 'Viscosity' and 'Surface Tension'; state 'Pascal's Law' and its application in hydraulic jacks.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□□ □□□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□ □□□□□□□□□□□ □□□□□ □□□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□□.

Common mistake / precaution: High pressure in hydraulic systems can be dangerous. Do not operate or maintain pressurized systems without authorised safety training and pressure-rated equipment.

2. Hydrodynamics Fundamentals−

Fluid dynamics studies fluids in motion. The 'Continuity Equation' is based on the law of conservation of mass. 'Bernoulli's Theorem' relates pressure, velocity, and elevation in a flowing fluid, serving as the basis for flow measurement.

Study and application method

State 'Bernoulli's Theorem' and its assumptions; explain the significance of the 'Continuity Equation' in pipe flow.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: State 'Bernoulli's Theorem' and its assumptions; explain the significance of the 'Continuity Equation' in pipe flow.

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Common mistake / precaution: Bernoulli's theorem assumes ideal (non-viscous) fluid. Do not apply it to real fluid flows without authorised correction factors for head loss.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Flow Through Pipes and Open Channels

Flow through orifices and mouthpieces; Flow through pipes: Darcy-Weisbach equation, Major and minor losses; Open channel flow: Chezy's and Manning's equations, Most economical sections.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

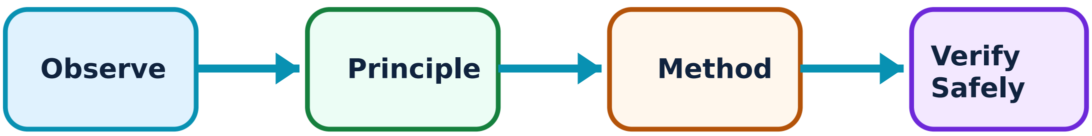
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Flow Through Pipes and Open Channels: identify the system, state its principle, follow the correct method and verify the result safely.

1. Pipe Flow and Losses−

Water flowing in pipes experiences resistance, leading to 'Head Loss'. Major loss is due to friction (Darcy-Weisbach equation), while minor losses occur due to bends, valves, and changes in cross-section.

Study and application method

Calculate the 'Frictional Head Loss' in a pipe using Darcy's formula; list three types of 'Minor Losses' in pipe systems.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Calculate the 'Frictional Head Loss' in a pipe using Darcy's formula; list three types of 'Minor Losses' in pipe systems.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Excessive head loss reduces system efficiency. Do not design pipe networks without authorised consideration of friction factors and optimal pipe diameters.

2. Open Channel Hydraulics−

Open channel flow has a free surface exposed to the atmosphere (e.g., canals, rivers). Flow is driven by gravity. The 'Chezy' and 'Manning' equations are used to calculate velocity. 'Most Economical Sections' are those that provide maximum discharge for a given area.

Study and application method

Compare 'Pipe Flow' and 'Open Channel Flow' conceptually; describe the 'Rectangular' and 'Trapezoidal' most economical sections.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare 'Pipe Flow' and 'Open Channel Flow' conceptually; describe the 'Rectangular' and 'Trapezoidal' most economical sections.

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Common mistake / precaution: Erosion and silting can damage canals. All canal designs must follow authorised 'Non-silting' and 'Non-scouring' velocity limits.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Irrigation Engineering and Canals

Methods of irrigation: Surface, Sub-surface, Sprinkler, Drip; Crop water requirements: Delta, Duty, Base period; Canal systems: Alignment, Lining, Cross-drainage works.

Approx. 16 hours | CO3 | Understand · Apply

Learning goals

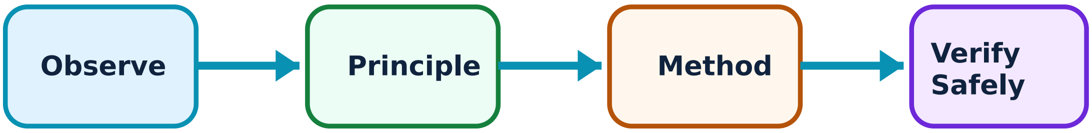
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Irrigation Engineering and Canals: identify the system, state its principle, follow the correct method and verify the result safely.

1. Irrigation Methods and Requirements−

Irrigation is the artificial application of water to crops. Surface irrigation is common but less efficient. Drip and Sprinkler irrigation save water. 'Duty' is the area irrigated per unit discharge, while 'Delta' is the total depth of water required by a crop.

Study and application method

Define 'Duty' and 'Delta' and state their relationship; explain the advantages of 'Drip Irrigation' for water conservation.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define 'Duty' and 'Delta' and state their relationship; explain the advantages of 'Drip Irrigation' for water conservation.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□ □□□□□□□□□□□ □□□□□ □□□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Common mistake / precaution: Over-irrigation can lead to waterlogging and salinity. Do not apply water without authorised soil-moisture analysis and crop-specific water scheduling.

2. Canal Networks and Structures−

Canals transport water from sources to fields. Lining reduces seepage losses. 'Cross-drainage Works' (Aqueducts, Super-passages) are required when a canal crosses a natural stream.

Study and application method

Differentiate between an 'Aqueduct' and a 'Super-passage' conceptually; explain why 'Canal Lining' is important.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Differentiate between an 'Aqueduct' and a 'Super-passage' conceptually; explain why 'Canal Lining' is important.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Seepage from unlined canals can raise the water table. All canal projects must follow authorised lining specifications and drainage plans.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Dams and Water Resource Management

Gravity dams: Forces, Stability, Failure modes; Spillways and Gates; River training works: Spurs, Groynes, Embankments; Rainwater harvesting and Watershed management.

Approx. 14 hours | CO4 | Understand · Apply

Learning goals

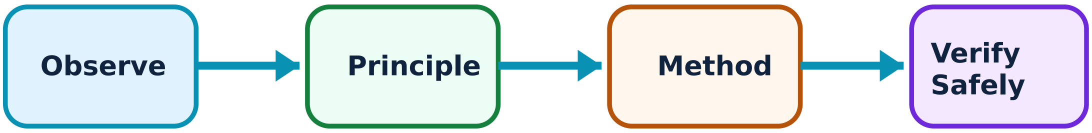
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Dams and Water Resource Management: identify the system, state its principle, follow the correct method and verify the result safely.

1. Gravity Dams and Spillways−

A gravity dam resists external forces (water pressure, uplift) by its own weight. Stability analysis involves checking against sliding and overturning. Spillways are 'safety valves' that release excess water during floods.

Study and application method

List the 'Forces' acting on a gravity dam; explain the purpose of a 'Spillway' and its gates.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List the 'Forces' acting on a gravity dam; explain the purpose of a 'Spillway' and its gates.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Dam failure is catastrophic. All dam designs must follow authorised seismic analysis, geological surveys, and emergency action plans (EAP).

2. Water Management and River Training−

River training works control the flow and protect banks from erosion. Watershed management involves the sustainable use of land and water resources. Rainwater harvesting is critical for groundwater recharge in water-stressed areas.

Study and application method

Describe the role of 'Spurs' in river training; explain the concept of 'Watershed Management' conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the role of 'Spurs' in river training; explain the concept of 'Watershed Management' conceptually.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Improper river training can shift erosion to downstream areas. All river interventions must follow authorised environmental impact assessments (EIA) and hydraulic modeling.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Fluid Properties and Hydrostatics

Use the concept flow to revise observation, principle, method and verification.

Module 2: Flow Through Pipes and Open Channels

Use the concept flow to revise observation, principle, method and verification.

Module 3: Irrigation Engineering and Canals

Use the concept flow to revise observation, principle, method and verification.

Module 4: Dams and Water Resource Management

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare Surface and Drip irrigation in a conceptual table showing efficiency and cost differences.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch a 'Trapezoidal Canal Section' showing the freeboard and lining.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw the 'Total Pressure' distribution on a vertical rectangular surface conceptually.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate the 'Duty' of water conceptually for a given area and discharge.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify a local irrigation structure (e.g., small check dam, canal sluice) and note its function.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Fluid Properties and Hydrostatics

Explain Fluid Properties and Pressure and state one correct method, application or precaution.−

Fluids (liquids and gases) deform continuously under shear stress. Key properties include density, viscosity (resistance to flow), and surface tension. Hydrostatics deals with fluids at rest. Pascal's law states that pressure at any point in a fluid at rest is equal in all directions.

Answer framework: Define 'Viscosity' and 'Surface Tension'; state 'Pascal's Law' and its application in hydraulic jacks.

Important point: High pressure in hydraulic systems can be dangerous. Do not operate or maintain pressurized systems without authorised safety training and pressure-rated equipment.

Explain Hydrodynamics Fundamentals and state one correct method, application or precaution.−

Fluid dynamics studies fluids in motion. The 'Continuity Equation' is based on the law of conservation of mass. 'Bernoulli's Theorem' relates pressure, velocity, and elevation in a flowing fluid, serving as the basis for flow measurement.

Answer framework: State 'Bernoulli's Theorem' and its assumptions; explain the significance of the 'Continuity Equation' in pipe flow.

Important point: Bernoulli's theorem assumes ideal (non-viscous) fluid. Do not apply it to real fluid flows without authorised correction factors for head loss.

Module 2 — Flow Through Pipes and Open Channels

Explain Pipe Flow and Losses and state one correct method, application or precaution.−

Water flowing in pipes experiences resistance, leading to 'Head Loss'. Major loss is due to friction (Darcy-Weisbach equation), while minor losses occur due to bends, valves, and changes in cross-section.

Answer framework: Calculate the 'Frictional Head Loss' in a pipe using Darcy's formula; list three types of 'Minor Losses' in pipe systems.

Important point: Excessive head loss reduces system efficiency. Do not design pipe networks without authorised consideration of friction factors and optimal pipe diameters.

Explain Open Channel Hydraulics and state one correct method, application or

precaution.−

Open channel flow has a free surface exposed to the atmosphere (e.g., canals, rivers). Flow is driven by gravity. The 'Chezy' and 'Manning' equations are used to calculate velocity. 'Most Economical Sections' are those that provide maximum discharge for a given area.

Answer framework: Compare 'Pipe Flow' and 'Open Channel Flow' conceptually; describe the 'Rectangular' and 'Trapezoidal' most economical sections.

Important point: Erosion and silting can damage canals. All canal designs must follow authorised 'Non-silting' and 'Non-scouring' velocity limits.

Module 3 — Irrigation Engineering and Canals

Explain Irrigation Methods and Requirements and state one correct method, application or precaution.−

Irrigation is the artificial application of water to crops. Surface irrigation is common but less efficient. Drip and Sprinkler irrigation save water. 'Duty' is the area irrigated per unit discharge, while 'Delta' is the total depth of water required by a crop.

Answer framework: Define 'Duty' and 'Delta' and state their relationship; explain the advantages of 'Drip Irrigation' for water conservation.

Important point: Over-irrigation can lead to waterlogging and salinity. Do not apply water without authorised soil-moisture analysis and crop-specific water scheduling.

Explain Canal Networks and Structures and state one correct method, application or precaution.−

Canals transport water from sources to fields. Lining reduces seepage losses. 'Cross-drainage Works' (Aqueducts, Super-passages) are required when a canal crosses a natural stream.

Answer framework: Differentiate between an 'Aqueduct' and a 'Super-passage' conceptually; explain why 'Canal Lining' is important.

Important point: Seepage from unlined canals can raise the water table. All canal projects must follow authorised lining specifications and drainage plans.

Module 4 — Dams and Water Resource Management

Explain Gravity Dams and Spillways and state one correct method, application or precaution.−

A gravity dam resists external forces (water pressure, uplift) by its own weight. Stability analysis involves checking against sliding and overturning. Spillways are 'safety valves' that release excess water during floods.

Answer framework: List the 'Forces' acting on a gravity dam; explain the purpose of a 'Spillway' and its gates.

Important point: Dam failure is catastrophic. All dam designs must follow authorised seismic analysis, geological surveys, and emergency action plans (EAP).

Explain Water Management and River Training and state one correct method, application or precaution.−

River training works control the flow and protect banks from erosion. Watershed management involves the sustainable use of land and water resources. Rainwater harvesting is critical for groundwater recharge in water-stressed areas.

Answer framework: Describe the role of 'Spurs' in river training; explain the concept of 'Watershed Management' conceptually.

Important point: Improper river training can shift erosion to downstream areas. All river interventions must follow authorised environmental impact assessments (EIA) and hydraulic modeling.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Fluid Properties and Hydrostatics.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Flow Through Pipes and Open Channels.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Irrigation Engineering and Canals.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Dams and Water Resource Management.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Fluid properties: Density, Viscosity, Surface tension; Hydrostatic pressure: Pascal's law, Total pressure, Center of pressure; Buoyancy and Floatation; Fundamentals of fluid flow: Continuity equation, Bernoulli's theorem..
2. Develop a complete answer or practical plan for: Flow through orifices and mouthpieces; Flow through pipes: Darcy-Weisbach equation, Major and minor losses; Open channel flow: Chezy's and Manning's equations, Most economical sections..
3. Develop a complete answer or practical plan for: Methods of irrigation: Surface, Sub-surface, Sprinkler, Drip; Crop water requirements: Delta, Duty, Base period; Canal systems: Alignment, Lining, Cross-drainage works..
4. Develop a complete answer or practical plan for: Gravity dams: Forces, Stability, Failure modes; Spillways and Gates; River training works: Spurs, Groynes, Embankments; Rainwater harvesting and Watershed management..

LAST-MILE REVISION

Quick Revision

Module 1: Fluid Properties and Hydrostatics

Fluid properties: Density, Viscosity, Surface tension; Hydrostatic pressure: Pascal's law, Total pressure, Center of pressure; Buoyancy and Floatation; Fundamentals of fluid flow: Continuity equation, Bernoulli's theorem.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Flow Through Pipes and Open Channels

Flow through orifices and mouthpieces; Flow through pipes: Darcy-Weisbach equation, Major and minor losses; Open channel flow: Chezy's and Manning's equations, Most economical sections.

- Match each topic to CO2.

- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Irrigation Engineering and Canals

Methods of irrigation: Surface, Sub-surface, Sprinkler, Drip; Crop water requirements: Delta, Duty, Base period; Canal systems: Alignment, Lining, Cross-drainage works.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Dams and Water Resource Management

Gravity dams: Forces, Stability, Failure modes; Spillways and Gates; River training works: Spurs, Groynes, Embankments; Rainwater harvesting and Watershed management.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Fluid Properties and Pressure	Fluids (liquids and gases) deform continuously under shear stress.
Hydrodynamics Fundamentals	Fluid dynamics studies fluids in motion.
Pipe Flow and Losses	Water flowing in pipes experiences resistance, leading to 'Head Loss'.
Open Channel Hydraulics	Open channel flow has a free surface exposed to the atmosphere (e.
Irrigation Methods and Requirements	Irrigation is the artificial application of water to crops.
Canal Networks and Structures	Canals transport water from sources to fields.
Gravity Dams and Spillways	A gravity dam resists external forces (water pressure, uplift) by its own weight.
Water Management and River Training	River training works control the flow and protect banks from erosion.

LEARNING RESOURCES

References

Recommended hydraulics and irrigation references

1. S. K. Garg, Irrigation Engineering and Hydraulic Structures.
2. B. C. Punmia and Pande B. B. Lal, Irrigation and Water Power Engineering.
3. K. R. Arora, Irrigation, Water Power and Water Resources Engineering.
4. P. N. Modi and S. M. Seth, Hydraulics and Fluid Mechanics.

Approved public hydraulics and irrigation sources

- https://nptel.ac.in/courses/105105203
- https://nptel.ac.in/courses/112103249

These sources provide the verified study basis while official Course 5013 detail is unavailable; they do not replace official dam designs, authorised irrigation projects, certified equipment specifications or authorised flood control plans.